A Few Caveats

Progress on coding and math is largely due to how easy those are for a computer to **verify**.

There's no reason to believe models will improve as fast in **other domains**.

AI Agents

What Is an Agent?

A chatbot **answers**. An agent **acts**.



Where Agents Shine, and Where They Don’t

GOOD AT

- Tedious, repetitive chores
- Tasks with a clear “done” check
- Working across files, browser, and code
- Reading a lot, quickly

STRUGGLES WITH

- Vague goals
- Rare, specialized facts
- Judgment calls
- Mistakes that can’t be undone

Things You Could Ask an Agent

SIMPLE	“Rename every photo in my Field Samples folder to date_site_specimen.jpg, using each photo’s date and my sample log spreadsheet.”
BROWSER	“Go to the NSF website, find open programs that fund undergraduate research in my field, and make a table of deadlines, award sizes, and eligibility.”
RESEARCH	“Find recent papers on [my topic], compare their methods and findings with my draft, flag where my results agree or disagree, and propose three testable hypotheses with an experiment for each.”

Science Relies On...

Data analysis and literature review	Easy for models
Proposing hypotheses	Easy, but unreliable
Physical experiments	Hard for models

Timelines in Science Are Long

1543

Copernicus: Sun-centered model

Late 1500s

Brahe: precise planetary observations

1609

Kepler: elliptical orbits

1610

Galileo: phases of Venus

1687

Newton: universal gravitation

1838

Bessel: stellar parallax

Where AI Is Moving Fast in Science

MATHEMATICS

80-year-old Erdős conjecture **disproved**

OpenAI model · May 2026

WEATHER

3-day hurricane forecasts as good as the old **1-day**

Google DeepMind WeatherNext · used by the National Hurricane Center

DRUG DISCOVERY

AI-chosen target, AI-designed drug, now in **Phase III**

Insilico's rentosertib, lung fibrosis · July 2026

MATERIALS

120 experiments in 12 hours, run by a robot lab

NC State's PoLARIS · lead-free nanomaterials · May 2026